Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and
Communications Technology Department.

Academic Year: 2022-2023
Semester: Spring
Exam Date: 7 / 6 / 2023
Level: 4th Level

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELC 422) Digital Signal Processing
Spec.: Communications & Computer

Examiners: Dr. Samir M. Kamal and Dr. Sara M. Hassan
Time: 3 Hours

Number of Pages: One | Number of Questions: 4 | Answer All Questions

Question 1 [ILO's: a1, a3, b2, b3, c2,c4] **(20 points)**

a- In A/D, in detail, explain the quantization process of continuous-amplitude signals. 3

b- Show that the DFT, X(K), of a sequence x(n) is periodic. 2

-> c- For the following sequence: x(n) = {1, 0, 1, 0}:

(i) Using the FFT algorithm, find the frequency components of this sequence. 4

->(ii) Check your results obtained in part (i) by calculating x(n) using the IDFT method. 4

(iii) Using your results, verify Parseval's theorem. 2.5

(iv) Plot the amplitude and the phase spectrums. 2.5

(v) What is the computational saving if we use FFT algorithm instead of DFT algorithm. 2

Question 2 [ILO's: a1,a2,b2,c1,c2] **(15 points)**

a- What are the different steps of the digital filter design? 2.5

b- Define (using equations and/or figures) the following items: Correlation and Convolution operations. 2.5

c - Given two sets of samples $x_1(n) = \{3, 2, 0, 1\}$ and $x_2(n) = \{1, 2, 0, 1\}$.

(i) Find the linear cross- correlation between $x_1(n)$ and $x_2(n)$ and plot it v.s. the lag number, j. 4+1

(ii) Find their convolution and plot it v.s. n. 4+1

Question 3 [ILO's: a4,b3,b4,c2,c3] **(15 points)**

a- For LTIS define (using equations and/or figures): Stability, Linearity and ROC. 3

b- For the following transfer function: $H(z) = z^2 (z+0.5) / (z-0.5)*(z-1)^2$

(i) Plot the pole - zero diagram of H (z). 1.5

(ii) Discuss the stability of this system. 1.5

(iii) Find the difference equation of this filter, y (n). 3

(iv) Represent the filtering operation in a block diagram form, 2

c- For H(z) given in (b), using the partial fraction (Residual) method find "4" values for the impulse response of this system. 4

Question 4 [ILO's: a1,a2,b2,c1,c2] **(10 points)**

a - In detail, what are the main characteristics of the discrete time sinusoidal signals? 3

b- For the signal $x(t) = \cos(200\pi t)$ to be sampled with sampling frequency $F_s = 400$Hz.

(i) Find an expression for its corresponding sequence, x(n). Then calculate its fundamental period NP. 3

(ii) Find the samples values of the first cycle in x(n), obtained in part (i), then plot x(n) and x(t) at the same figure. 4

Good Luck